QAI DATA CENTER AS A SERVICE (QAI-DCaaS)

Client and Partner Concept Proposal

A stage-gated model aligning investment, engineering, operations, assurance, and commercial outcomes.

Proposal status

This is an initial discussion proposal for prospective clients, investors, infrastructure providers, technology vendors, engineering partners, research organizations, and public-sector stakeholders. It describes a proposed service model, not a binding offer, fixed-price quote, certification claim, guaranteed delivery commitment or promise of investment returns. Scope, schedule, pricing, contractual allocation, and assurance targets will be finalized by stakeholders after discovery and feasibility review.

1. Executive summary

Bhadale IT proposes to develop QAI Data Center as a Service (QAI-DCaaS): a coordinated service that helps clients and partners turn a datacenter or hybrid classical–quantum infrastructure ambition into an assessable, engineered, governed and commercially testable programme.

Rather than requiring a client to coordinate disconnected contractors, technology suppliers, facility specialists, software teams, security advisers, financiers, and operators independently, the model establishes a common plan, decision gates, responsibilities, evidence requirements, and transparent cost assumptions. Specialist providers remain accountable for their contracted products and services; Bhadale IT's proposed role is to integrate the plan, manage interfaces and balance stakeholder expectations.

- Start with needs, constraints, use cases and existing assets—not a predetermined hardware purchase.
- Use virtual-first validation and classical baselines before committing to physical infrastructure.
- Make investment decisions through explicit gates with evidence, accountable approvers, documented risks, and go/no-go outcomes.
- Separate technical design, deployment control, managed operations, and independent certification.
- Test commercial viability through lifecycle costing, demand validation, staged funding, measurable outcomes, and sensitivity analysis.

2. Proposed service model

QAI-DCaaS is the client-facing coordination and delivery model. It can cover feasibility, architecture, programme planning, engineering integration, deployment management, service readiness, lifecycle economics, and continuous improvement. The package may range from a bounded assessment to a multi-year programme, depending on client needs and partner capability.

Component	Purpose	Boundary
QAI Data Center Blueprint	Defines target architecture, principles, deployment patterns, and interfaces.	Architecture reference: does not itself deploy or certify a facility.
QAI Deployment Operator	Supports resource/workload assessment, deployment planning, approvals, execution boundaries, and evidence capture.	Software capability under staged validation; not a substitute for physical facility controls.
QAI-DCaaS	Packages investment planning, engineering, PMO coordination, supplier integration, and lifecycle oversight.	Actual scope depends on agreed work packages and qualified partners.
Data Center Operations Service	May provide recurring monitoring, support, maintenance coordination, and service reporting.	Separate staffing, tools, SLAs, and liabilities must be agreed.
Independent assurance	External assessment, audit, or certification under a selected scheme.	Performed by a competent independent body; not self-awarded by Bhadale IT.

3. Stakeholder roles and collaboration

The client appoints an accountable sponsor and retains approval authority for investment, risk appetite, scope, and acceptance. Bhadale IT can function as an integrator and program/engineering coordination lead where contracted. Specialists retain responsibility for professional decisions and deliverables within their contracts.

Stakeholder	Contribution / decision rights	Collaboration evidence
Client / asset owner	Defines outcomes, constraints, site/data needs, risk appetite, and acceptance criteria; approves scope, budget and stage exits.	Requirements, decisions, acceptance and change approvals.
Investor / HoldCo / finance partner	Sets investment criteria, funding envelope, return/risk expectations, and funding conditions.	Capital assumptions, staged releases, downside/sensitivity scenarios.
Bhadale IT — proposed integrator	Coordinates discovery, architecture alignment, work packages, dependencies, risks, gates, change control, and reporting.	Integrated plan, RACI, RAID and decision logs, evidence matrix, dashboards, and escalation.
Facility / colocation provider	Confirms site capacity, physical security, power, cooling, access, resilience, and facility operating rules.	Site survey, capacity evidence, constraints, and procedures.
Compute / storage / network / GPU / FPGA suppliers	Provide specifications, prices, support, lifecycle, and integration requirements.	Qualified configurations, quotations, compatibility, and warranty terms.
QPU / quantum control / QCaaS provider	Provides modality-specific constraints, access/API terms, usage charges, and telemetry where available.	Backend capabilities, availability limitations, data/IP terms, and dependencies.
Systems / software / integration partners	Build approved interfaces, workflows, observability, security, and operations components.	Design artifacts, interface contracts, test reports and releases.
Security, privacy, legal and compliance specialists	Assess data handling, cyber controls, sovereignty, safety, regulatory and assurance needs.	Risk/control assessments, contract reviews, remediation and scoped sign-off.
Operations / service management	Defines support, incident/change/problem processes, monitoring, maintenance, continuity, and reporting.	Runbooks, RACI, SLA/SLO proposals, escalation, and recovery tests.
Independent assessor / certification body	Assesses the defined scope against the chosen independent scheme.	Formal findings, corrective actions, and certification decision if applicable.
End users / product owner / domain experts	Validate workflows, usability, workload value and operational fit.	User stories, baseline measures, acceptance tests, and feedback.
Public-sector / research / community stakeholders, where relevant	Clarify public needs, research collaboration, local capability, environmental and community considerations.	Requirements, consultation, procurement compliance, and agreed outcomes.

4. End-to-end stage-gate lifecycle

Each gate has an entry package, review participants, exit criteria and a recorded decision. A gate may approve, approve with conditions, request rework, defer pending evidence, or stop the programme. Tailor the sequence to scale, jurisdiction, risk and procurement model.

Stage / gate	Main activities	Decision and minimum evidence
G0 — Opportunity and fit	Clarify client problems, outcomes, stakeholders, urgency, and strategic fit.	Proceed / no-bid / defer; problem statement, sponsor, initial scope, assumptions, and conflict check.
G1 — Discovery and asset assessment	Inventory systems, IP, workloads, data, facilities, skills, contracts, and constraints; classify maturity.	Approve findings; asset register, use cases, current-state map, risks, and known unknowns.
G2 — Feasibility and business case	Compare delivery options, demand assumptions, architecture, costs, dependencies, and risks.	Go / revise / stop; options analysis, preliminary CAPEX/OPEX/TCO, demand evidence, and sensitivity scenarios.
G3 — Architecture and assurance design	Define architecture, interfaces, data/security, facility assumptions, service boundaries, and standards applicability.	Approve baseline design; requirements/evidence matrix, threat/risk assessment, and design decisions.
G4 — Investment and programme authorization	Fund work packages; set milestones, roles, procurement route, schedule, contingency, and terms.	Authorize / conditionally authorize / reject; budget, cash-flow scenarios, RACI, acceptance criteria and contracts/LOIs as needed.
G5 — Detailed engineering and readiness	Complete designs, supplier selections, integration/test plans, operating procedures, and recovery plans.	Authorize build/pilot; reviewed designs, supplier commitments, readiness tests, and change baseline.
G6 — Build, integrate and verify	Implement in virtual/sandbox environment first where feasible; test functionality, security, performance, and resilience.	Accept or remediate; traceable test evidence, defect log, configuration record, and known limitations.

G7 — Pilot / commissioning / service acceptance	Run bounded workloads; compare baseline; test monitoring, incident handling, recovery, and support.	Accept / extend / remediate / stop; pilot report, residual risks, operational readiness, and acceptance evidence.
G8 — Independent assurance and launch approval	Complete applicable external assessments; resolve findings; approve catalogue, pricing, support, and disclosures.	Launch only within approved scope; assessment reports if relevant, closure evidence and approved claims.
G9 — Operate, review and scale	Monitor service, costs, utilization, outcomes, incidents, risks, and suppliers; assess expansion or retirement.	Continue / optimize / scale / pause / retire; KPI/SLA reports, benefit realization, and updated business case.

5. Feedback, change control, and approval discipline

- Maintain one requirements and decision register: source, priority, owner, acceptance method, and status for every material requirement.
- Hold scheduled design and stage reviews with client, integrator and relevant technical, facility, security, finance, and operations representatives.
- Record feedback, assess impact on scope/cost/schedule/security/operations, assign a decision owner, and close the loop with evidence.
- Require written approval for material changes to scope, budget, interfaces, security posture, or acceptance criteria.
- Record gate outcomes: approve; approve with conditions and dates; rework; hold pending evidence; or stop. Include rationale and unresolved risks.
- Escalate unresolved cross-party conflicts with the client sponsor or steering committee. Bhadale IT facilitates options and impact analysis but does not override client or regulator authority.
- Maintain traceability: Need → requirement → architecture decision → work package → implementation → test → approval → service metric → business outcome.

6. How Bhadale IT balances expectations

Potential tension	Balancing approach	Decision principle
Investor return vs engineering risk	Stage-gated funding, ranges, and sensitivities; do not commit all capital before technical and demand evidence.	Release capital against evidence and risk limits.
Client scope vs budget/schedule	Prioritize outcomes; separate must-have from optional scope; expose trade-offs and baseline changes.	No hidden scope expansion.
Supplier claims vs interoperability	Vendor-neutral requirements where practical; test interfaces; document dependencies and lock-in.	Select on evidence, lifecycle cost, fit and support.
Innovation vs readiness	Virtual-first, classical baselines, and QPU/hardware only where justified.	No unsupported production-readiness or quantum-advantage claims.
Speed vs security/compliance	Integrate reviews early; time-box remediation without bypassing mandatory controls.	Risk acceptance rests with authorized client owner, subject to law and contract.
Availability promises vs supplier dependency	Define boundaries, exclusions, maintenance, and upstream dependencies before SLAs.	Commit only to measurable and controllable targets.
Public value vs private return	Define public outcomes, procurement integrity, accountability, and IP/data terms.	Disclose conflicts; separate public obligations from investor assumptions.
Reuse vs client-specific needs	Maintain reusable core blueprint and controlled client configurations.	Do not misrepresent a tailored design as certified.

Bhadale IT's value is integration discipline: helping stakeholders see the same plan, understand dependencies and trade-offs, and make decisions from shared evidence. This does not replace the client sponsor, licensed engineers, facility operators, legal advisers, security authorities, suppliers or independent assessors.

7. Commercial model and route to profitability

Profitability is a target to evaluate, not a guaranteed result. Build the business case from explicit demand, scope, delivery capacity, third-party costs, risks, and pricing assumptions. A multi-year programme should not rely on speculative returns or unverified quantum advantage.

Commercial element	Possible approach	Controls needed
Discovery / feasibility	Fixed fee for bounded assessment, or capped time-and-materials.	Defined inputs, deliverables, duration, and exclusions.
Blueprint / engineering	Milestone-based professional services.	Acceptance criteria, revision allowance, client decisions, and IP terms.
Programme / PMO / integration	Monthly retainer, named work packages or milestone fees.	Capacity assumptions, governance scope, reporting, and change control.
Deployment / integration	Milestone/work-package pricing; hardware and third parties separately quoted.	Quotes, contingency, acceptance tests, warranty, and dependencies.
Managed operations	Recurring service fee with optional usage-based components.	Service boundary, staffing, support hours, monitoring and SLAs/SLOs.
QPU / cloud / facilities / licences	Pass-through, separately contracted, or bundled only if cost/risk understood.	Usage limits, price changes, data terms, availability, and renewal.
Optimization / expansion	Change programme or outcome-linked fee where measurable.	Baseline, measurement, attribution, and supportable savings claims.

7.1 Financial model inputs

- Revenue: discovery, blueprint milestones, program management, integration of work packages, recurring operations, and optional optimization.
- Direct costs: engineering, specialist partners, hardware, cloud/QPU usage, facility/colocation, licenses, security, testing, travel, and external assurance.
- Indirect costs: sales/bid effort, overhead, legal/accounting, insurance, administration, platform development, and support readiness.
- Risk allowance: contingency, FX, supply lead times, scope change, performance uncertainty, customer delays, and supplier dependencies.
- Scenarios: conservative/base/upside demand, utilization ramp, delivery capacity, renewals/churn, cost escalation, cash timing and working capital.
- Outputs: gross/contribution margin by offer, break-even revenue/utilization, cash flow, and scenario-based ROI/NPV/IRR only where inputs are credible.

No numeric price, revenue forecast or return rate is asserted in this document. Develop these after discovery, partner quotations, capacity planning, and demand validation.

8. Assurance, service quality, and standards pathway

Embed assurance in engineering and operations. Distinguish (1) standards-informed design, (2) verified operational controls, and (3) independent assessment/certification. These are separate outcomes and must not be claimed before scope and evidence exist.

- Software/deployment: schema and interface validation, reproducible tests, authorization, compatibility, failure handling, evidence capture and release/change control.
- Service management: catalogue, responsibilities, incident/problem/change handling, maintenance, monitoring, customer reporting, continuity, and supplier management.
- Security/data governance: identity/access, tenant boundaries, sensitive workload metadata, data location/retention, logging, incident response, and supplier responsibilities.
- Facility assurance: site, physical security, power, cooling, fire/life safety, resilience, commissioning, maintenance, and sustainability metrics as applicable.
- Candidate references for applicability review: ISO/IEC 22237 / EN 50600, ISO/IEC 27001, ISO 22301 and ISO/IEC 20000-1, plus relevant local laws, codes, and customer requirements.
- For relevant Australian large data-centre proposals, assess government expectations on national interest, energy transition, water, workforce, research, and local capability.
- Choose any independent assessment route with competent specialists once service scope and target market are clear.

9. Indicative delivery roadmap

This is a planning envelope, not a committed schedule. Timing depends on scope, decisions, engineering capacity, procurement lead times, site readiness, partner agreements, and assurance requirements.

Indicative period	Focus	Outputs / exit evidence
Weeks 0–4	Discovery, stakeholder alignment, asset, and use-case inventory.	Agreed problem, sponsor, stakeholder map, initial risks and discovery report.
Weeks 4–8	Feasibility, options, preliminary architecture, and commercial scenarios.	Options paper, high-level blueprint, cost ranges, assumptions, and gate decision.
Weeks 8–14	Detailed planning, service requirements, assurance matrix, and work packages.	Architecture baseline, RACI, delivery plan, funding gates, and acceptance plan.
Weeks 14–26	Virtual proof, deployment lifecycle validation, and integration spikes.	Demonstration, test evidence, gap list, revised costs/schedule, and pilot recommendation.
Months 6–12+	Partner/hardware integration and bounded pilot, if approved/funded.	Pilot report, service-readiness review, updated operating model and go/no-go.
Year 2 onward, gated	Scale, operations, independent assessment, and commercialization.	Service catalogue, operational KPIs, assurance outcomes and reviewed business case.

10. Deliverables

- Stakeholder map, governance charter, RACI and decision/escalation paths.
- Current-state asset/capability assessment, use cases and maturity register.
- Options analysis, reference architecture, service boundaries, and interface/dependency map.
- Requirements, risk and assurance matrix with verification methods and evidence owners.
- Work breakdown structure, work-package descriptions, schedule, resource plan, and gate criteria.
- Commercial model with transparent assumptions, CAPEX/OPEX/TCO categories, and scenario analysis.
- Deployment, test, security, operations and service-readiness plan appropriate to approved scope.
- Pilot/commissioning report, acceptance record, limitations, corrective actions, and next-stage recommendations.
- Service catalogue and draft operating/SLA/SLO model if recurring service delivery is included.

11. Responsibilities, assumptions, and exclusions

11.1 Responsibilities

- Client: appoint sponsor/decision owners; provide accurate requirements, asset/site/data information, and timely approvals; fund approved packages; accept/reject deliverables.
- Bhadale IT: coordinate agreed work, maintain plans/registers, facilitate decisions, manage dependencies, and report status/risks; deliver only contracted work packages.
- Partners/suppliers: provide specifications, quotations, licences, warranties, service commitments, professional deliverables, and support for their scope.
- Independent specialists: provide advice, assessment, or certification only within their engagement and competence.
- All parties: agree IP, confidentiality, data, security, conflicts, change control, acceptance, and dispute-resolution terms before delivery.

11.2 Assumptions and exclusions

- This proposal assumes staged discovery and approval; it does not authorize procurement, construction, production deployment, or third-party commitments.
- No fixed price, financing commitment, investment return, availability, certification, quantum advantage, or production-readiness claim is offered.
- Facilities, hardware, QPU access, cloud, licences, site works, independent audits/certification, legal advice, and specialist engineering are separately scoped unless expressly included.
- Schedule and cost depend on client inputs, approvals, partner availability, site readiness, procurement, and scope stability.
- Background IP remains with its owner; foreground IP, licensing, reuse, confidentiality, and publication rights require contract terms.
- Regulatory, safety, legal, tax, financing, and certification decisions require qualified professionals and relevant authorities.

12. Key risks and mitigations

Risk	Mitigation / gate control
Scope expands before feasibility	Time-box discovery; prioritize outcomes; formal change control.
Costs/returns rely on weak assumptions	Show sources/confidence; use sensitivities; update at every gate.
Hardware selected before workload needs	Virtual proof and profiling first; vendor-neutral requirements and procurement gate.
Supplier dependency/lock-in	Document interfaces, exit options, portability, support, and lifecycle costs.
Security/assurance discovered late	Review security and standards at architecture and detailed-design gates.
Service levels exceed control	Define boundaries and dependencies before setting targets.
Stakeholders disagree	Use decision rights, evidence packs, steering escalation, and recorded approvals.
Demand does not materialize	Validate demand before major capex; stage funding and stop/go decisions.

13. Governance and reporting

- Weekly delivery review: work-package progress, blockers, decisions, changes, defects, and next actions.
- Monthly steering review: scope, schedule, cost, risks, partner dependencies, benefits, and decisions needed.
- Each gate: concise evidence pack, decision record, conditions, accountable approver, and next-stage funding recommendation.
- Quarterly service/commercial review during operations: customer outcomes, costs, margins, service metrics, incidents, supplier performance, and scale/retire options.

14. Proposed next step

Start with a bounded discovery and feasibility engagement. Agree the target use case, stakeholders, existing assets, service boundary, success criteria, data/site constraints, partner needs, decision rights, and budget envelope. The output is a client-reviewed options report and recommendation to proceed, revise or stop before larger commitments.

After discovery, the parties can decide whether to commission a blueprint, virtual proof, Deployment Operator validation package, partner-led infrastructure design or broader QAI-DCaaS programme.

15. Reference context

- **Data Center Dynamics — Are data centers ready for quantum in the cloud?** https://www.datacenterdynamics.com/en/opinions/are-data-centers-ready-for-quantum-in-the-cloud/
- **arXiv — Towards A High-Performance Quantum Data Center Network Architecture:** https://arxiv.org/pdf/2509.09653
- **arXiv — Quantum Computing as a Service: a Software Engineering Perspective:** https://arxiv.org/html/2510.04982v1
- **Australian Government — Expectations of data centres and AI infrastructure developers:** https://www.industry.gov.au/publications/expectations-data-centres-and-ai-infrastructure-developers
- **ASHRAE — AI Data Center Energy Performance Framework:** https://www.ashrae.org/technical-resources/topics-and-initiatives/ai-data-center/introduction-and-purpose
- **IDCA — Data Center Standards:** https://www.idc-a.org/data-center-standards
- **GovLoop / ViON — Data Center as-a-Service:** https://go.govloop.com/rs/231-DWB-776/images/Data-Center-as-a%20Service.pdf
- **Data Center Asia — What is Data Center as a Service?** https://www.datacenter-asia.com/blog/what-is-data-center-as-a-service-dcaas/

16. Closing statement

QAI-DCaaS is proposed as a disciplined way to bring client outcomes, investment, engineering, suppliers, service operations, and assurance into one governed lifecycle. Its value is not a promise that every risk disappears or every investment becomes profitable. Its value is that risks, costs, responsibilities, assumptions, and evidence become visible early enough for stakeholders to make informed decisions, correct courses and stop when evidence does not support proceeding. For further details please email: vijaymohire@bhadaleit.com